



Sleep Disorders (PDQ®)–Health Professional Version

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Overview

Insomnia symptoms occur in about 33% to 50% of the adult population [1] and are often associated with situational stress, illness, aging, and drug treatment.[2] It is estimated that one-third to one-half of people with cancer experience sleep disturbances.[3,4] Physical illness, pain, hospitalization, drugs and other treatments for cancer, and the psychological impact of a malignant disease may disrupt the sleeping patterns of people with cancer.[5] Adequate sleep may increase the cancer patient's pain tolerance. Poor sleep adversely affects daytime mood and performance. In the general population, persistent insomnia has been associated with a higher risk of developing clinical anxiety or depression.[6] Sleep disturbances and, ultimately, sleep-wake cycle reversals can be early signs of a developing delirium.

Sleep consists of two phases:[7]

1. Rapid eye movement (REM) sleep: REM sleep, also known as dream sleep, is the active or paradoxical phase of sleep in which the brain is active.
2. Non-REM (NREM) sleep: NREM sleep is the quiet or restful phase of sleep. NREM, also referred to as slow-wave sleep, is divided into four stages of progressively deepening sleep based on electroencephalogram findings.

Questions?

The stages of sleep occur in a repeated pattern or cycle of NREM followed by REM, with each cycle lasting approximately 90 minutes. The sleep cycle is repeated four to six times during a 7- to 8-hour sleep period.[7] The sleep-wake cycle is dictated by an inherent biological clock or circadian rhythm. Disruptions in individual sleep patterns can disrupt the circadian rhythm and impair the sleep cycle.[8]

The Sleep Disorders Classification Committee of the American Academy of Sleep Medicine has identified five major categories of sleep disorders:[9]

1. Disorders of initiating and maintaining sleep (insomnias).
2. Sleep-related breathing disorders (sleep apnea).
3. Disorders of excessive somnolence (hypersomnias).
4. Disorders of the sleep-wake cycle (circadian rhythm sleep disorders).
5. Dysfunctions associated with sleep, sleep stages, or partial arousals (parasomnias).

In this summary, unless otherwise stated, evidence and practice issues as they relate to adults are discussed. The evidence and application to practice related to children may differ significantly from information related to adults. When specific information about the care of children is available, it is summarized under its own heading.

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Sleep Disturbances in Cancer Patients

Cancer patients are at great risk of developing insomnia and disorders of the sleep-wake cycle. Insomnia, the most common sleep disturbance in this population, is most often secondary to physical and/or psychological factors related to cancer and/or cancer treatment.[1-6] Anxiety and depression—common psychological responses to the diagnosis of cancer, cancer treatment, and hospitalization—are highly correlated with insomnia.[7,8];[9][Level of evidence: II] In addition, sleep disturbances may be part of cancer-related symptom clusters. For more information, see [Symptom Clusters in Cancer](#).

Sleep disturbances may be exacerbated by paraneoplastic syndromes associated with steroid production and by symptoms associated with tumor invasion, such as:

- Draining lesions.
- Gastrointestinal (GI) and genitourinary (GU) alterations.
- Pain.
- Fever.
- Cough.
- Dyspnea.
- Pruritus.
- Fatigue.

Sleep disturbance can also vary by diagnosis. In a study of patients with melanoma (n = 124), breast cancer (n = 124), and endometrial cancer (n = 82),[10][Level of evidence: II] symptom profiles differed by diagnosis. Four symptom profiles were identified: minimally symptomatic, insomnia-predominant, very sleepy with upper airway symptoms, and symptomatic with severe dysfunction. Using latent class profile analysis, group differences by cancer diagnosis were identified, as shown in Table 1:

Table 1. Sleep-Disturbance Symptom Profiles by Cancer Diagnosis

Cancer Type	Minimally Symptomatic	Insomnia-Predominant	Very Sleepy With Upper Airway Symptoms	Symptomatic With Severe Dysfunction
Melanoma	X		X	
Breast		X		X
Endometrial	X	X	X	X

Differences by disease groups may be associated with treatment regimens and/or other factors.

Medications—including vitamins, corticosteroids, neuroleptics for nausea and vomiting, and sympathomimetics for the treatment of dyspnea—and other treatment factors can negatively impact sleep patterns.

Side effects of treatment that may affect the sleep-wake cycle include the following:[11];[9][Level of evidence: II]

- Pain.
- Anxiety.
- Night sweats/hot flashes. For more information, see [Hot Flashes and Night Sweats](#).
- GI disturbances (e.g., incontinence, diarrhea, constipation, or nausea).
- GU disturbances (e.g., incontinence, retention, or GU irritation).
- Respiratory disturbances.
- Fatigue.

Sustained use of the following can cause insomnia:

- Sedatives and hypnotics (e.g., glutethimide, benzodiazepines, pentobarbital, chloral hydrate, secobarbital sodium, and amobarbital sodium).
- Anticonvulsants (e.g., phenytoin).
- Corticosteroids.
- Oral contraceptives.
- Monoamine oxidase inhibitors.
- Methyldopa.
- Propranolol.
- Atenolol.
- Alcohol.
- Thyroid preparations.

In addition, withdrawal from the following substances may cause insomnia:

- Central nervous system depressants (e.g., barbiturates, opioids, glutethimide, chloral hydrate, methaqualone, ethchlorvynol, alcohol, and over-the-counter and prescription antihistamine sedatives).
- Benzodiazepines.

- Major tranquilizers.
- Tricyclic and monamine oxidase inhibitor antidepressants.
- Illicit drugs (e.g., marijuana, cocaine, phencyclidine, and opioids).

Hypnotics can interfere with rapid eye movement (REM) sleep, resulting in increased irritability, apathy, and diminished mental alertness. Abrupt withdrawal of hypnotics and sedatives may lead to symptoms such as:

- Nervousness.
- Jitteriness.
- Seizures.
- REM rebound.

REM rebound is a marked increase in REM sleep, with increased frequency and intensity of dreaming, including nightmares.[12] The increased physiological arousal that occurs during REM rebound may be dangerous for patients with peptic ulcers or a history of cardiovascular problems. Newer medications for insomnia have reduced adverse effects.[13]

The sleep of hospitalized patients is likely to be frequently interrupted by treatment schedules, hospital routines, and roommates, which singularly or collectively alter the sleep-wake cycle. Other factors influencing sleep-wake cycles in the hospital setting include patient age, comfort, pain, anxiety, environmental noise, and temperature.[14]

Consequences of sleep disturbances can influence outcomes of therapeutic and supportive care measures.[15] The patient with mild to moderate sleep disturbances may experience irritability and inability to concentrate, which may in turn affect compliance with treatment protocols, the ability to make decisions, and relationships with significant others. Sleep disturbances can also cause depression and anxiety. Supportive care measures are directed toward promoting quality of life and adequate rest.

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Assessment

Assessment is the initial step in managing sleep disturbances in people with cancer. Assessment data should include the following:[1]

- Documentation of predisposing factors.
- Sleep patterns.
- Emotional status.
- Exercise and activity levels.
- Diet.
- Symptoms.
- Medications.
- Caregiver routines.

The sections below outline recommendations for a sleep history and physical examination. Data can be retrieved from multiple sources, such as:[2]

- The patient's subjective report of sleep difficulty.
- Objective observations of behavioral and physiological manifestations of sleep disturbances.
- Reports from the patient's significant other regarding the patient's quality of sleep.

The Insomnia Severity Index, which has been validated in adult oncology populations, is recommended when screening for insomnia in clinical settings.[3,4] In a 2021 study, the Insomnia Severity Index was validated in young adult (18–40 years of age) cancer survivors.[5]

The diagnosis of insomnia is primarily based on a careful, detailed medical and psychiatric history. The American Academy of Sleep Medicine has produced guidelines for the use of polysomnography as an objective tool in evaluating insomnia. The routine polysomnogram includes the monitoring of the following:

- Electroencephalography.

- Electro-oculography.
- Electromyography.
- Effort of breathing and air flow.
- Oxygen saturation.
- Electrocardiography.
- Body position.

Polysomnography is the major diagnostic tool for assessment of sleep disorders. It is indicated in the evaluation of suspected sleep-related breathing disorders and periodic limb movement disorder, when the cause of insomnia is uncertain, or when behavioral or pharmacological therapy is unsuccessful.[6]

Sleep disturbances have been shown to change throughout the cancer trajectory, which supports the need to assess sleep throughout the patient's cancer experience. One descriptive study [7][[Level of evidence: II](#)] involving 398 women with breast cancer used the General Sleep Disturbance Scale (GSDS) to identify three different sleep trajectories when self-reported sleep was evaluated beginning before surgery and continuing for 6 months. One group (55% of the sample) had a high level of sleep disturbance throughout the study, defined as scores on the GSDS of about 58 to 60 at all data points. A second group (40% of the sample) was considered to have a low level of sleep disturbance throughout, defined as scores on the GSDS in the low 30s at each data point. The final group (5% of the sample) started out high, with scores around 62, but their scores decreased to below 30 over the first 4 months and remained there through month 6. Participants identified as having a more severe sleep disorder were significantly younger, had more comorbidities, had a lower performance status, and experienced hot flashes. In another study, of 232 women with gynecologic cancers that assessed sleep using the GSDS at six time points over two cycles of chemotherapy, four distinct subgroups of patients with sleep disturbance were identified (Low, 18.5%; Moderate, 43.6%; High, 29.3%; Very High, 8.6%).[8][[Level of evidence: II](#)] Participants with the worst sleep disturbance were younger, had a higher body mass index, and were more likely to report depression or back pain.

Sleep disturbances frequently co-occur with cancer-related fatigue and may have common underlying mechanisms. Prospective and multidimensional assessments of these two symptoms were conducted in a study involving patients who were newly diagnosed with stage I through stage IIIA breast cancer (N = 152).[9] Assessments included validated sleep and fatigue questionnaires and objective sleep assessments using wrist actigraphy, which measure sleep-wake patterns and circadian rhythms. Assessments were conducted before initiation of chemotherapy (T1) and during the last week of the fourth cycle of anthracycline-based chemotherapy (T2). Most patients in the group characterized by severe symptoms at T1 were also in a higher-symptoms group at T2. Similarly, many patients in that group at T1 remained in the minimal-symptoms group at T2. From T1 to T2, the average-symptoms group was relatively unstable compared with the severe- and minimal-symptoms groups. At both time points, younger patients reported more severe symptoms, and married patients reported less severe symptoms. Patients who reported more comorbidities, more use of medications, and other indicators of worse health (e.g., higher BMI) were more likely to be in the group characterized by higher symptom severity at both time points.

In addition, stress can contribute to sleep disturbance. A study of 957 patients undergoing chemotherapy treatment for breast, lung, gastrointestinal, or gynecological cancer measured responses to validated stress/resilience assessment instruments (14-item Perceived Stress Scale, 22-item Impact of Event Scale-Revised, 30-item Life Stressor Checklist-Revised, and 10-item Connor-Davidson Resilience Scale). Compared with patients classified in a normative or resilient class, patients in the stressed class had significantly higher levels of sleep disturbance. Also, each of the domains of sleep disturbance within the validated 21-item GSDS (quantity, quality, sleep onset latency, mid-sleep awakening, early awakenings, and excessive daytime sleepiness) and use of medications to help with sleep were statistically significant and crossed clinically meaningful cutoff points for patients in the stressed class.[10]

Risk Factors for Sleep Disorders

- Disease factors, including paraneoplastic syndromes with increased steroid production, and symptoms associated with tumor invasion (e.g., obstruction, pain, fever, shortness of breath, pruritus, and fatigue).
- Treatment factors, including symptoms related to surgery (e.g., pain, frequent monitoring, and use of opioids), chemotherapy (e.g., exogenous corticosteroids), and symptoms related to chemotherapy.
- Medications such as opioids; sedatives/hypnotics; steroids; caffeine/nicotine; some antidepressants; and dietary supplements, including some vitamins, diet pills, and other products promoting weight loss and appetite suppression.
- Environmental factors.
- Physical and/or psychological stressors.
- Depression. For more information, see [Depression](#).
- Anxiety. For more information, see the [Anxiety Disorders: Description and Etiology](#) section in Adjustment to Cancer: Anxiety and Distress.
- Delirium.
- Daytime seizures, snoring, and headaches.

Characterization of Sleep

- Usual patterns of sleep, including usual bedtime, routine before retiring (e.g., food, bath, and medications), length of time before onset of sleep, and duration of sleep (waking episodes during night, ability to resume sleep, and usual time of awakening).
- Characteristics of disturbed sleep (changes after diagnosis, treatment, and/or hospitalization).
- The significant other's perception of the patient's quantity and quality of sleep.
- Family history of sleep disorders.

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Management

Management of sleep disturbances should focus on treatment of problems such as:

- Falling asleep.
- Staying asleep.
- Early-morning awakenings.

Other areas to manage include symptoms from cancer and its treatment and the identification and management of environmental and psychological factors. When sleep disturbances are caused by symptoms of cancer or its treatment, measures that control or alleviate symptoms are often the key to resolving sleep disturbances. Management of sleep disturbances combines nonpharmacological and pharmacological approaches individualized for the patient.

Nonpharmacological Management of Sleep Disturbances

Many people who experience insomnia have poor sleep hygiene (such as smoking and drinking excessive alcohol just before bedtime), which can exacerbate or perpetuate insomnia.[1][[Level of evidence: III](#)] A complete assessment of sleep hygiene (i.e., time in bed; napping during the day; intake of caffeine, alcohol, or foods that are heavy, spicy, or sugary; exercise; and sleep environment) and use of behavioral management strategies (i.e., fixed bedtime; restrictions on smoking, diet, and excessive alcohol 4–6 hours before bedtime; and increased exercise) may prove effective in reducing sleep disturbances.

Sleep hygiene in an inpatient setting involves modifying the sleep environment to decrease sleep disruption. Minimizing noise, dimming or turning off lights, adjusting room temperature, and consolidating patient care tasks to reduce the number of interruptions can increase the amount of uninterrupted sleep.[2][[Level of evidence: IV](#)]

Cognitive strategies include:[3]

- Restructuring negative thoughts, beliefs, and attitudes related to sleep.
- Preventing excessive monitoring or worrying about getting enough sleep.

Behavioral strategies include:

- Stimulus control.
- Sleep restriction.

Both of these strategies seek to limit the time spent in bed that does not involve sleeping.[3-5] Several large randomized trials and meta-analyses provide the evidence base for the efficacy of cognitive behavioral therapy (CBT) for insomnia (CBT-I).[3,6,7] Most of these trials have been in populations of patients without cancer.

Components of CBT-I include the following:

- Cognitive restructuring.
- Behavioral strategies.
- Relaxation training.[4]
- Basic sleep hygiene education.

Relaxation therapy can be used to achieve both behavioral and cognitive outcomes, particularly when it is combined with imagery. Educational objectives around sleep hygiene are also used to treat insomnia and include content on the following:[4]

- Sleeping and waking up at regular times.
- Relaxing before bedtime.
- Creating a dark, comfortable sleep environment.
- Avoiding watching television or working in the bedroom.
- Getting ample daylight during nonsleep hours.
- Avoiding naps.
- Limiting caffeine.
- Getting regular exercise but no closer than 3 hours before bedtime.

Practice guidelines from the American Academy of Sleep Medicine clearly state that multicomponent therapy is recommended over single therapies. Because of insufficient evidence about its efficacy, sleep hygiene education is not recommended as a single-modality management approach; other reviews state that sleep hygiene by itself is not effective.[6,8] Information about sleep hygiene should be included in patient education about sleep issues.

Several trials and meta-analyses have shown CBT-I to be at least as effective as conventional pharmacological therapies in treating primary chronic insomnia but without side effects.[6,7,9-11]

A four-arm study (conducted in patients with primary chronic insomnia) that evaluated zolpidem versus CBT versus zolpidem and CBT versus placebo reported a greater effect ($P = .05$) on sleep-onset latency for both groups involving CBT (change of 44%) versus the group receiving zolpidem alone (change of 29%).[12] Another study, also conducted in patients with primary chronic insomnia, evaluated CBT with temazepam alone versus a combination of CBT and temazepam versus placebo and found that all active treatments were significantly better than placebo and that there was a trend for the most improvement in the combined arm of CBT and temazepam.[13] Both arms with CBT demonstrated greater reductions in time to sleep onset than the pharmacotherapy-alone arm (64% in the combined arm, 55% in the CBT arm, and 47% in the temazepam arm). A meta-analysis examining pharmacological and behavioral studies for persistent insomnia found that pharmacological and behavioral treatments did not differ in magnitude of benefit except for latency to sleep onset, in which greater reductions were found with behavioral therapy.[7]

There are limited data evaluating elements of CBT-I in cancer survivors, and most data are about women with breast cancer. However, there have been at least four randomized controlled trials of CBT-I in cancer survivors.[14-17] The intervention was typically delivered over five to eight weekly, small-group, in-person sessions. One trial included patients with cancer diagnoses other than breast cancer,[16] and results did not differ by cancer diagnosis. All studies showed improvements in numerous sleep parameters over time in the groups receiving CBT-I and demonstrated continued benefits 6 and 12 months later. Two of the four trials did not use active control arms.[14,16]

Most studies using active control arms were in breast cancer survivors. One study compared CBT-I with sleep education and hygiene in 72 women,[15] while the other study used a healthy-eating education control group.[17] In the first study, both groups significantly improved over time, with some significant differences between groups favoring CBT-I for time to fall asleep, time awake after sleep, total sleep time, and overall sleep quality. For example, the group receiving CBT-I improved by 30 minutes in time to fall asleep, compared with 11 minutes in the sleep education and hygiene group.[15]

In the second study, 219 women were randomly assigned to a behavioral therapy group consisting of stimulus control, general sleep hygiene (limiting naps, going to bed and rising at consistent times), and relaxation or to a healthy-eating education control group. The interventions were delivered by trained nurses in person, 2 days before the initiation of

chemotherapy, before each chemotherapy treatment, and 30 days after the last chemotherapy treatment. The nurses worked with women assigned to behavioral therapy to individualize and reinforce the behaviors. The Pittsburgh Sleep Quality Index (PSQI) was used to measure subjective sleep quality, complemented by use of a sleep diary and wrist actigraph. Sleep quality significantly improved in the group receiving behavioral therapy, compared with the control group. These differences were also seen in data from the sleep diary and actigraph, with both showing significantly fewer awakenings in the behavioral therapy group.[18] Sleep quality was significantly better at 90 days and at 1 year in the behavioral therapy group, as measured by the PSQI but not the diary or actigraph.[17]

In some places, patients may not have access to in-person, professionally delivered CBT-I. A randomized controlled trial conducted with breast cancer survivors demonstrated that CBT-I delivered via digital media can also produce meaningful clinical improvements, although improvements are not as robust as those produced with professionally delivered CBT-I. This three-armed trial compared video-based CBT-I (VCBT-I) and professionally delivered CBT-I (PCBT-I) with a no-treatment control group in 242 breast cancer survivors. Both the VCBT-I and PCBT-I groups had significantly greater improvements in diary-measured sleep variables, compared with the control group. The patients in the PCBT-I group reported greater improvements in some sleep outcomes and in fatigue and depression levels than did the VCBT-I group.[19]

Table 2. Evidence for Cognitive Behavioral Therapy for Insomnia (CBT-I) in Cancer Survivors

Reference	Cancer Type	Sample Size and Design	Control and CBT-I Intervention	Measures	Outcomes
Berger et al., 2009 [17]	Breast (stages I–III) during chemotherapy	N = 219; RCT	Control: Healthy-eating group (sessions with equal time, attention)	PSQI, sleep diary, actigraphy ^a , fatigue assessment	Significant improvement in sleep quality and nighttime awakenings for CBT group, compared with control group
			CBT-I: Individualized plan before chemotherapy, stimulus control, modified sleep restriction, relaxation therapy, sleep hygiene		
Epstein et al., 2007 [15]	Breast (stages I–III)	N = 72; RCT	Control: Sleep education and hygiene	Sleep diary, actigraphy, ISI	Both groups improved over time; significant improvement between groups favored CBT-I group in time to fall asleep, time awake
			CBT-I: 6 sessions,		

ISI = Insomnia Severity Index; PCBT-I = professionally administered CBT-I; PSQI = Pittsburgh Sleep Quality Index; QOL = quality of life; RCT = randomized controlled trial; VCBT-I = video-based CBT-I.

^aActigraphy: A technique that uses a small instrument called an actigraph (a watch-like sensory unit) worn on the wrist or ankle to measure body gross motor activity. It is helpful in determining sleep patterns and daytime activity.

^bPolysomnography: A test used to diagnose sleep disorders on the basis of sleep-related biophysiological changes.

Reference	Cancer Type	Sample Size and Design	Control and CBT-I Intervention	Measures	Outcomes
			stimulus control, sleep restriction, sleep education and hygiene		after sleep onset, total sleep time, sleep quality (as measured by ISI)
Espie et al., 2008 [16]	Mixed	N = 150; RCT	Control: Sleep education and hygiene	Sleep diary; actigraphy; fatigue, depression/anxiety, and QOL assessments	Significant improvement in time to fall asleep, time awake after sleep onset, sleep efficiency, fatigue, specific QOL outcomes for CBT-I group, compared with control group
			CBT-I: 5 weekly sessions, stimulus control, sleep restriction, cognitive restructuring		
Savard et al., 2005 [14]	Breast (stages I–III)	N = 57; RCT	Control: Wait list	Sleep diary; polysomnography ^b ; ISI; fatigue, depression/anxiety, and QOL assessments	Significant improvement in time to fall asleep, time awake after sleep onset, sleep efficiency, depression/anxiety, and QOL outcomes for CBT group, compared with control group
			CBT-I: 8 weekly sessions, stimulus control, sleep restriction, sleep education and hygiene, cognitive restructuring, fatigue management		
Savard et al., 2014 [19]	Breast (stages I–III)	N = 242; RCT	Control: No treatment (n = 81)	Sleep diary; ISI; actigraphy; fatigue, depression/anxiety, and QOL assessments	Compared with control group, PCBT-I and VCBT-I groups associated with significant improvement in sleep diary-measured sleep variables; compared with VCBT-I group, PCBT-I group had more improvement in sleep, fatigue, and depression/anxiety outcomes and had higher remission rates for insomnia
			PCBT-I (n = 81): 6 weekly sessions		
			VCBT-I (n = 80): 60-min animated video, 6 booklets		
			CBT-I content: Similar for both		

ISI = Insomnia Severity Index; PCBT-I = professionally administered CBT-I; PSQI = Pittsburgh Sleep Quality Index; QOL = quality of life; RCT = randomized controlled trial; VCBT-I = video-based CBT-I.

^aActigraphy: A technique that uses a small instrument called an actigraph (a watch-like sensory unit) worn on the wrist or ankle to measure body gross motor activity. It is helpful in determining sleep patterns and daytime activity.

^bPolysomnography: A test used to diagnose sleep disorders on the basis of sleep-related biophysiological changes.

Reference	Cancer Type	Sample Size and Design	Control and CBT-I Intervention	Measures	Outcomes
			groups (stimulus control, sleep restriction, sleep education and hygiene, cognitive restructuring)		

ISI = Insomnia Severity Index; PCBT-I = professionally administered CBT-I; PSQI = Pittsburgh Sleep Quality Index; QOL = quality of life; RCT = randomized controlled trial; VCBT-I = video-based CBT-I.

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^bPolysomnography: A test used to diagnose sleep disorders on the basis of sleep-related biophysiological changes.

Inpatient nonpharmacological management

CBT delivered by psychologists has shown promise for the treatment of insomnia in patients with cancer.[14][[Level of evidence: I](#)] A randomized controlled study investigated the effectiveness of a protocol-driven cognitive behavioral intervention for insomnia delivered by oncology nurses.[16][[Level of evidence: I](#)] This group intervention consisted of standard CBT components such as stimulus control and sleep restriction. Participants included patients with heterogeneous cancers randomly assigned to receive the intervention (n = 100) or treatment as usual (n = 50). Primary outcomes were sleep diary measures at baseline, posttreatment, and at 6-month follow-up. CBT was associated with significant and sustained improvements in several sleep aspects. These improvements were seen for both subjective (sleep diary) and objective (actigraphy) assessments. Additionally, patients who received CBT showed significant improvements in fatigue, anxiety, and depressive symptoms and reported improved quality of life, compared with patients who received treatment as usual.[16][[Level of evidence: I](#)]

A study conducted in cancer survivors demonstrated the benefits of a specialized yoga program to improve sleep quality and reduce medication use. A total of 410 cancer survivors with moderate to severe sleep disturbances were randomly assigned to receive standard care or standard care plus a 4-week yoga intervention delivered in two weekly sessions by trained yoga instructors. The yoga participants showed significant improvement in sleep quality, daytime dysfunction, nighttime awakening, and sleep efficiency, compared with standard-care participants. One major limitation of this study was its limited population generalizability, as most study participants were female, White, married, and well-educated breast cancer survivors. Another major limitation was the lack of an adequate control group with respect to nonspecific effects such as group support and attention.[20]

Exercise interventions have shown positive effects on subjective and objective sleep quality in patients with cancer. A study conducted in Taiwanese patients with lung cancer investigated the effects of a 12-week exercise intervention on sleep quality and rest-activity rhythms.[21] The intervention included a home-based walking exercise regimen (walking at a moderate intensity for 40 minutes, 3 times per week) and weekly exercise counseling. Participants were randomly assigned to either the intervention group (n = 56) or the usual-care group (n = 55). Assessments conducted at baseline, 3 months, and 6 months included a subjective sleep assessment using the PSQI and objective sleep and rest-activity assessment using actigraphy.

Over time, the walking exercise group showed significant improvement in subjective sleep quality (lower PSQI scores) compared with the usual-care group. The walking exercise group also showed improvement in total sleep time (TST), an important objective measure of sleep quality, compared with the usual-care group. Additionally, the positive effects on TST (i.e., increase in TST) were greater in patients with poor rest-activity rhythm at baseline, suggesting more benefits in patients with poor circadian sleep-activity rhythms. The limitations of the study included a lack of blinding, hence a

possible placebo effect in the intervention group. Also, despite randomization, the mean amount of baseline moderate physical activity was higher in the usual-care group than in the intervention group.

Other actions or interventions that may promote rest in the hospital or extended-care setting include the following: [22,23]

- Keeping the patient's skin clean and dry.
- Giving back rubs and/or massaging areas of the body to bring comfort to the patient (e.g., bony prominences, head and scalp, shoulders, hands, and feet).
- Keeping bedding and/or surfaces of support devices (chairs and pillows) clean, dry, and wrinkle-free.
- Ensuring adequate bedcovers for warmth.
- Regulating fluid intake to avoid frequent awakening for elimination.
- Encouraging bowel and bladder elimination before sleep.
- Promoting optimal bowel function (increased fluids, dietary fiber, and use of stool softeners and laxatives).
- Using a condom catheter for nocturnal incontinence.
- Providing a high-protein snack 2 hours before bedtime (e.g., milk, turkey, or other foods high in tryptophan).
- Avoiding beverages with caffeine and other stimulants, including dietary supplements that promote metabolism changes and appetite suppression.
- Encouraging the patient to dress in loose, soft clothing.
- Facilitating comfort through repositioning and support with pillows as needed.
- Encouraging activity and being out of bed as much as possible during waking hours.
- Encouraging the patient to keep regular bedtime and waking hours and avoid napping during the day.
- Minimizing and coordinating necessary bedside contacts.

Psychological interventions are directed toward facilitating the patient's coping processes through education, support, and reassurance. As the patient learns to cope with the stresses of illness, hospitalization, and treatment, sleep may improve.[24][[Level of evidence: IV](#)] Communication, verbalization of concerns, and openness between the patient, family, and health care team should be encouraged. Relaxation exercises and self-hypnosis performed at bedtime can help promote calm and sleep. Cognitive-behavioral interventions that diminish the distress associated with early insomnia and change the goal from "need to sleep" to "just relax" can diminish anxiety and promote sleep.[25]

Pharmacological Management of Sleep-Wake Cycle Disturbances

When cancer survivors experience sleep-wake disturbances, cognitive behavioral intervention counseling should be the first consideration for management. For more information, see the [Nonpharmacological Management of Sleep Disturbances](#) section. Resources for education and training in CBT may not be readily available in many cancer centers; therefore, community resources need to be investigated. If CBT is not available or has not been successful, pharmacological management can be considered. In addition, when patients have comorbidities contributing to sleep-wake cycle disturbances (such as hot flashes, uncontrolled pain, anxiety, depression, or other mood disturbances), [26,27] then pharmacological management will probably be necessary. While many pharmacological agents are approved for primary insomnia and many others are used off-label to manage sleep and related symptoms, most of the approved sleep aids have not been studied in cancer populations; therefore, the risk/benefit profiles of these drugs are not delineated in this setting.

Despite the lack of evidence in cancer populations, clinicians widely use pharmacological interventions. Therefore, the following discussion of pharmacological agents and recommendations for use is based on evidence from studies conducted in patients with primary insomnia and clinical experience.[4,28,29]

Several classes of medications are used to treat sleep-wake cycle disturbances, including the following:

- Nonbenzodiazepine benzodiazepine receptor agonists (e.g., zolpidem).
- Benzodiazepines (e.g., lorazepam).
- Melatonin receptor agonists (e.g., ramelteon).
- Antihistamines (e.g., hydroxyzine).
- Antidepressants (e.g., trazodone) and antipsychotics (e.g., quetiapine) that have sedative effects.

Drug characteristics to consider before a drug is chosen to treat an individual patient include the following:

- Absorption.
- Time to reach maximum concentration.
- Elimination half-life.
- Receptor activity.
- Ability to cross the blood-brain barrier.
- Dose and frequency.
- Formulation (short-acting versus long-acting).

These pharmacokinetic principles are important to match the agent to the type of sleep disturbance (e.g., problems falling asleep versus problems staying asleep). There are also safety issues to be considered, such as potentials for tolerance, abuse, dependence, withdrawal (including risk of rebound insomnia), and drug-drug and drug-disease interactions. Medications for sleep-wake cycle disturbances should be used short term and/or as needed.

General considerations for the use of hypnotics

Medications used to induce sleep are intended for the short-term management of sleep disorders. The use of these medications for longer periods is poorly studied. They are usually combined with lifestyle changes that reinforce good sleep habits and negate the need for chronic hypnotic medications.

Most research studies of current and historic hypnotic medications rarely exceed a duration of 12 to 16 weeks. Additionally, no current hypnotic re-creates normal sleep architecture, and variations from normal periods of rapid eye movement (REM) sleep and non-REM sleep are common. It is important to taper hypnotic medications slowly, or the variations in normal sleep patterns can become even more pronounced, with the majority of time spent in REM sleep in a condition known as REM rebound.[30,31]

Table 3 lists the drug categories and specific medications, including doses, commonly used to treat sleep disturbances.

Table 3. Medications Commonly Used to Promote Sleep

Drug Category	Medication	Dose	Comments	Reference
Nonbenzodiazepine benzodiazepine receptor agonist	zaleplon (Sonata)	5–20 mg	Useful for problems falling asleep only.	[32][Level of evidence: I]
	zolpidem tartrate (Ambien)	5–10 mg	Useful for problems falling asleep only. Maximum suggested	[32][Level of evidence: I]
CR = controlled-release; FDA = U.S. Food and Drug Administration; REM = rapid eye movement.				

Drug Category	Medication	Dose	Comments	Reference
			dose for women: 5 mg.	
	zolpidem tartrate extended-release (Ambien CR)	6.25–12.5 mg	Biphasic release; useful for problems both falling asleep and staying asleep. Do not crush or split tablets. Maximum suggested dose for women: 6.25 mg.	[32][Level of evidence: I]
	eszopiclone (Lunesta)	1–3 mg	Useful for problems both falling asleep and staying asleep. Do not take with or right after meal.	[32][Level of evidence: I]
Benzodiazepine	clonazepam (Klonopin)	0.25–2 mg	Used for REM sleep disorder (not FDA approved).	[32][Level of evidence: III];[33]
	lorazepam (Ativan)	0.5–4 mg; dose >2 mg rare	Risk of loss of motor coordination, falls, and cognitive impairment.	[32][Level of evidence: I]
	temazepam (Restoril)	7.5–30 mg	Risk of loss of motor coordination, falls, and cognitive impairment.	[32][Level of evidence: II]
Melatonin receptor agonist	ramelteon (Rozerem)	8 mg	Useful for problems falling asleep only. Little negative effect on cognition, somnolence, motor coordination, or nausea.	[32][Level of evidence: I]
Antihistamine	diphenhydramine (Benadryl)	25–100 mg	Useful for problems falling asleep only. Anticholinergic side effects; increases delirium risk in older patients.	[32][Level of evidence: I]
	hydroxyzine (Vistaril, Atarax)	10–100 mg	Useful for problems falling asleep only. Anticholinergic side effects; increases delirium risk in older patients.	[34][Level of evidence: II]
CR = controlled-release; FDA = U.S. Food and Drug Administration; REM = rapid eye movement.				

Drug Category	Medication	Dose	Comments	Reference
Tricyclic antidepressant	doxepin (Silenor)	3–6 mg	Lower doses used for treatment of primary insomnia when antidepressant effect not needed. Risk of anticholinergic side effects and weight gain.	[32] [Level of evidence: I]
	amitriptyline (Elavil)	10–25 mg	Lower doses used for treatment of primary insomnia when antidepressant effect not needed. Risk of anticholinergic side effects and weight gain.	[35] [Level of evidence: II]
	nortriptyline (Pamelor)	10–50 mg	Risk of anticholinergic side effects and weight gain.	[36] [Level of evidence: III]
Second-generation antidepressant	trazodone (Desyrel)	25–100 mg	Risk of orthostatic hypotension and falls.	[37]
	mirtazapine (Remeron)	7.5–45 mg	If depression not a concern, 7.5–15 mg best for sleep, hot flashes, increased appetite, and less morning sedation. Risk of falls.	[35] [Level of evidence: III]
Antipsychotic	quetiapine (Seroquel)	25–100 mg	Risk of weight gain, metabolic syndrome, abnormal/involuntary movements; possible cardiovascular effects (e.g., prolonged QT interval). Generally not a preferred agent due to side effects.	[38] [Level of evidence: III]
Chloral derivative	chloral hydrate	500–1,000 mg	Used mainly for sleep maintenance. Risk of gastric irritation, dependence, and withdrawal. Lethal in overdose.	[32] [Level of evidence: I]
CR = controlled-release; FDA = U.S. Food and Drug Administration; REM = rapid eye movement.				

Nonbenzodiazepine benzodiazepine receptor agonists

All agents in this class are FDA approved for primary insomnia. These agents promote sleep by enhancing the effects of gamma-aminobutyric acid (GABA) at the GABA type A (GABAA) receptor. Unlike traditional benzodiazepines, these

agents preferentially target specific GABAA receptor subtypes. Zolpidem and zaleplon bind predominantly to the alpha-1 subtype of GABAA, and eszopiclone preferentially targets the alpha-3 receptor subtype. This selective receptor subtype targeting has both advantages and disadvantages. These agents have mainly hypnotic/sedative effects and lack the anxiolytic, anticonvulsant, and myorelaxant effects seen with benzodiazepines. Conversely, because of the selective receptor subtype targeting, these agents have fewer effects on cognitive and psychomotor function and carry less risk of tolerance, dependence, and withdrawal (especially physical withdrawal) than benzodiazepines.[4,28,29]

These agents may be preferred for use in patients with cancer when only hypnotic effects are desired and should be taken just before bedtime (or even in bed) because they enter the brain very quickly. Some of these agents (e.g., zaleplon) have a short elimination half-life. Because of their longer-lasting effects, zolpidem extended-release and eszopiclone are preferred in the treatment of difficulty staying asleep. However, these agents carry a higher risk of residual morning sedation and cognitive/motor impairments than do agents with shorter elimination half-lives (e.g., zaleplon and immediate-release zolpidem).

Benzodiazepines

Benzodiazepines target several GABAA receptor subtypes, including alpha-1, -2, -3, and -5, and work by enhancing GABA effects at these receptors. In addition to hypnotic/sedative effects, these agents also have anxiolytic, anticonvulsant, and myorelaxant effects. Benzodiazepines are preferred when other effects (such as antianxiety or muscle relaxation) are desirable with or without the hypnotic effects.[4,28,29]

Benzodiazepines carry a much higher risk of tolerance, dependence, and withdrawal than nonbenzodiazepine receptor agonists. Benzodiazepine withdrawal has been associated with the risk of seizures, delirium tremens, autonomic instability, and death. These agents should be used with extreme caution and close monitoring in patients with histories of significant substance use because of potential tolerance and dependence issues. Benzodiazepines have also been associated with cognitive impairment and difficulties with motor coordination.

Generally, benzodiazepines with longer half-lives (e.g., clonazepam) are associated with a higher risk of residual morning sedation and cognitive/motor impairments. Agents with shorter elimination half-lives (e.g., lorazepam) are generally preferred for short-term anxiolytic effects and difficulties falling asleep and in older patients. Agents with longer half-lives (e.g., clonazepam) are preferred for the treatment of persistent anxiety and difficulties falling and staying asleep. All benzodiazepines are associated with risk of respiratory depression and should be used with caution in patients with preexisting respiratory disorders.

Melatonin receptor agonists: Ramelteon and tasimelteon

Ramelteon and tasimelteon work by binding to the melatonin receptor types MT1 and MT2. Ramelteon is useful only for the treatment of difficulties falling asleep and does not have any other effects, such as anxiolytic or myorelaxant effects. Tasimelteon is indicated for use in circadian sleep disorder. These agents do not treat difficulties staying asleep but also carry much less risk of cognitive/motor impairments and dependence.[28,29,39]

Antihistamines

Diphenhydramine and hydroxyzine decrease arousal by blocking histamine receptors. Antihistamines are sold over the counter and are useful for treating difficulties in falling asleep only. There is limited evidence for the use of antihistamines to treat insomnia; these agents are used when traditional hypnotics or benzodiazepines are less suitable because of the risk of cross-dependence or other issues, such as vulnerability of a patient to addictions. The anticholinergic properties of antihistamines may also be beneficial in the treatment of nausea and vomiting. The sedative and anticholinergic properties of these agents increase the risk of delirium, especially in older patients.[28,29]

Antidepressants

Sedating antidepressants are considered first-line agents when insomnia is comorbid with depression/anxiety symptomatology. These drugs include tricyclic antidepressants (e.g., amitriptyline) and second-generation

antidepressants (e.g., mirtazapine). The sedating effects of tricyclic antidepressants are caused mainly by histamine receptor blockading and partially by blockading of 5-HT₂ and muscarinic receptors. The sedating effects of mirtazapine are caused by its blocking of 5-HT₂ and histamine receptors, while those of trazodone are caused by its blocking actions at the histamine, 5-HT, and noradrenaline receptors.[4,28,29] For more information, see the [Pharmacological Intervention](#) section in Depression.

Tricyclic antidepressants have a small therapeutic window and can be lethal in overdose, compared with second-generation antidepressants such as mirtazapine. Additionally, tricyclics carry other risks, such as weight gain, anticholinergic side effects, and cardiovascular side effects, and should be used under close supervision. These agents sometimes are used in low doses (see [Table 3](#)) as adjuncts to other antidepressants to treat insomnia comorbid with depression/anxiety. This helps to avoid the side effects associated with higher doses while delivering the needed sedating effects. Tricyclics can also boost appetite and may be the treatment of choice for insomnia in patients with comorbid cachexia. Certain tricyclics (amitriptyline and nortriptyline) can also be beneficial in the treatment of pain syndromes (e.g., neuropathic pain) and headaches when these issues are comorbid with insomnia. Low doses of antidepressants (subtherapeutic for depression) are frequently used to treat insomnia without any comorbidities.

Mirtazapine has appetite-stimulating and antiemetic properties in addition to sedating effects. It is frequently used in insomniac patients with depression (therapeutic dose for depression, 15–45 mg) or without depression (subtherapeutic dose for depression, 7.5–15 mg) with comorbid nausea or loss of appetite. In low doses, trazodone (50–100 mg) can promote sleep and is often combined with other antidepressants (e.g., fluoxetine 20 mg in the morning) in depressed patients with insomnia.

Antipsychotics

Antipsychotics such as quetiapine have sedating effects caused mainly by the blockade of histamine receptors. However, these agents should be considered as a last resort and as a short-term treatment because of their serious side-effect profile. The use of antipsychotics has been associated with the following:

- Weight gain.
- Metabolic syndrome.
- Diabetes.
- Cardiovascular risks.
- The risk of extrapyramidal side effects, including tardive dyskinesia.

Antipsychotics can be considered for treatment-refractory insomnia, especially with comorbid anxiety symptomatology. [28]

Chloral derivative: Chloral hydrate

Chloral hydrate has sleep-promoting effects resulting from its effects on GABA systems. It is associated with risk of withdrawal symptoms similar to those of benzodiazepines and with rapid development of tolerance. Additionally, chloral hydrate carries the risk of gastric irritation and multiple drug interactions, and it is lethal in overdose. Like antipsychotics, chloral hydrate is usually considered only in cases of treatment-refractory insomnia because of its serious side-effect profile and the availability of safer alternatives.[28]

Botanical/dietary supplements

Melatonin

Melatonin, a hormone produced by the pineal gland during the hours of darkness, plays a major role in the sleep-wake cycle and has been linked to the circadian rhythm. A review found that short-term use of melatonin appears to be safe; however, the studies were not conducted in the context of cancer therapy.[40] Adjuvant melatonin may also improve sleep disruption caused by drugs known to alter normal melatonin production (e.g., beta-blockers and

benzodiazepines).[41][[Level of evidence: IV](#)] However, a meta-analysis of 25 studies exploring the efficacy and safety of melatonin in managing secondary sleep disorders or sleep disorders accompanying sleep restriction found that melatonin was not effective in these conditions.[42]

Evidence suggests that circulating melatonin levels are significantly lower in physically healthy older people and in insomniacs than in age-matched control subjects. In view of these findings, melatonin replacement therapy may be beneficial in the initiation and maintenance of sleep in older patients.[43][[Level of evidence: II](#)] A slow-release formulation of melatonin is licensed in Europe and is approved as monotherapy for patients aged 55 years or older for the short-term treatment (up to 13 weeks) of primary insomnia characterized by poor-quality sleep. However, melatonin replacement as a treatment for insomnia has not been studied in older people with cancer. Ramelteon and tasimelteon work via the melatonin receptor system: ramelteon to support the initiation of sleep, and tasimelteon to correct circadian sleep disorder.

Melatonin may interact with certain chemotherapeutic regimens via the cytochrome P450 enzyme and other systems.[44] It may augment the effects of some chemotherapeutic agents metabolized via the enzyme CYP1A2 and may exert inhibitory effects on P-glycoprotein-mediated doxorubicin efflux.

Clinical studies in individuals with renal, breast, colon, lung, and brain cancer suggest that melatonin exerts anticancer effects in conjunction with chemotherapy and radiation therapy; however, evidence remains inconclusive.[45,46] All of the studies suggesting antitumor effects of melatonin have been conducted by the same group of investigators and were open label. Efforts by independent groups of investigators are under way to investigate these effects in carefully designed, randomized, blinded studies.[45] *In vitro* and animal studies have demonstrated the anticancer effects of exogenous melatonin, and lower melatonin levels are associated with tumor growth.[47] Human studies have yet to substantiate any causal or associative relationships.

***Cannabis* and cannabinoids**

No studies have been conducted to specifically evaluate the effects of *Cannabis* inhalation or other *Cannabis* products in patients with primary or secondary sleep disturbances. Limited data from *in vitro* studies, animal studies, and small populations of healthy individuals or chronic *Cannabis* users are beginning to elucidate some of the relationships among various neurotransmitters, the sleep-wake cycle, and related effects of *Cannabis* pharmacology.[48,49]

Cannabis-based medicines are under development as a treatment for chronic pain syndromes, including cancer-related pain. One such medication is nabiximols (Sativex), an oromucosal formulation (delta-9-tetrahydrocannabinol and cannabidiol mixed in a 1:1 ratio). Studies conducted with nabiximols, primarily focusing on pain syndromes, have shown improvement in subjective sleep quality when sleep was measured as a secondary outcome.[50] Comorbidities such as pain are common reasons for sleep disturbances. Concerns have been raised about the abuse and dependence potential of nabiximols, especially in the subpopulation of patients with histories of *Cannabis* use.[51] Nabiximols is approved in Canada for the treatment of central neuropathic pain in patients with multiple sclerosis. In the United States, it is only available for investigational use and is currently under investigation for the treatment of intractable cancer pain. For more information, see [Cannabis and Cannabinoids](#).

Current Clinical Trials

Use our [advanced clinical trial search](#) to find NCI-supported cancer clinical trials that are now enrolling patients. The search can be narrowed by location of the trial, type of treatment, name of the drug, and other criteria. [General information](#) about clinical trials is also available.

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Special Considerations

The Patient With Pain

Since enhanced pain control improves sleep, appropriate analgesics or nonpharmacological pain management should be administered before introducing sleep medications. Tricyclic antidepressants can be particularly useful for the treatment of insomnia in patients with neuropathic pain and depression. Patients on high-dose opioids for pain may be at increased risk for the development of delirium and organic mental disorders. Such patients may benefit from the use of low-dose neuroleptics as sleep agents (e.g., haloperidol 0.5–1 mg).

The Older Patient

Older patients frequently have insomnia due to age-related changes in sleep. The sleep cycle in this population is characterized by lighter sleep, more frequent awakenings, and less total sleep time. Anxiety, depression, loss of social support, and a diagnosis of cancer are contributory factors in sleep disturbances in older patients.^[1]

Sleep problems in older adults are so common that nearly one-half of all hypnotic prescriptions written are for people older than 65 years. Although normal aging affects sleep, the clinician should evaluate the many factors that cause insomnia, such as:^[2]

- Medical illness.
- Psychiatric illness.
- Dementia.
- Alcohol and/or polypharmacy.
- Restless legs syndrome.
- Periodic leg movements.

- Sleep apnea syndrome.

Nonpharmacological treatment of sleep disorders is the preferred initial management, with the use of medication when indicated and referral to a sleep disorder center when specialized care is necessary.[2]

Providing a regular schedule of meals, discouraging daytime naps, and encouraging physical activity may improve sleep. Hypnotic prescriptions for older patients must be adjusted for variations in metabolism, increased fat stores, and increased sensitivity. Dosages should be reduced by 30% to 50%. Problems associated with drug accumulation (especially flurazepam) must be weighed against the risks of more severe withdrawal or rebound effects associated with short-acting benzodiazepines. An alternate drug for older patients is chloral hydrate.[1]

Sleep Apnea After Mandibulectomy

Anterior mandibulectomy can result in the development of sleep apnea. All patients with head and neck tumors who have had extensive anterior oral cavity resection should be evaluated before decannulation of the tracheostomy tube. Subsequent flap and/or reconstruction of the lower jaw seems to prevent the development of sleep apnea. In contrast, facial sling suspension of the lower lip does not prevent the development of sleep apnea.[3] Assessment for symptoms and preparation for the appearance of symptoms in this population provide indications for interventions related to sleep apnea.

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Latest Updates to This Summary (07/17/2024)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Editorial changes were made to this summary.

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About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about the pathophysiology and treatment of sleep disorders. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

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Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

The lead reviewers for Sleep Disorders are:

- Marilyn J. Hammer, PhD, DC, RN, FAAN (Dana-Farber Cancer Institute)
- Jayesh Kamath, MD, PhD (University of Connecticut Health Center)

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Updated: July 17, 2024

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